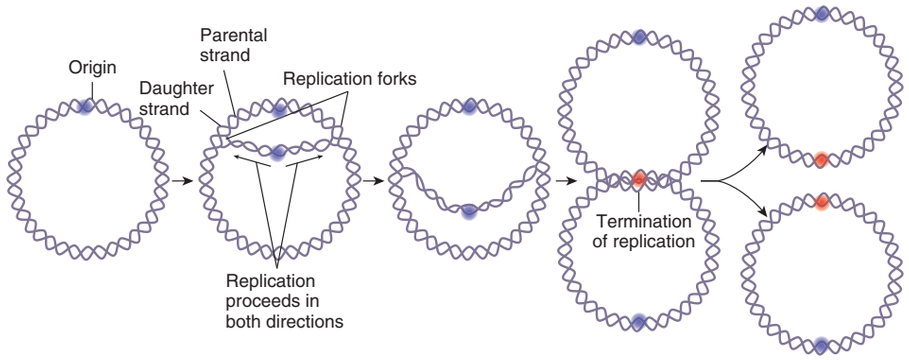


Figure 14.15: Termination of replication in a circular DNA

14.3.4 Eukaryotic DNA Replication

Replication in Eukaryotic cells is more complex. The DNA molecules in Eukaryotes are considerably larger than those in bacteria and are organized into complex nucleoprotein structures (chromatin). The essential features of DNA replication are same in eukaryotes and prokaryotes. However, some interesting variations do occur. Initiation of replication in all eukaryotes requires a multisubunit protein. Multiple origins of replication are probably a universal feature in eukaryotic cells. Like bacteria,

Infobits

DNA molecules exists in circular form in prokaryotic microorganisms, viruses and in organelles of eukaryotic organisms. However not all DNA molecules are circular. The chromosomes of eukaryotic organisms and of many viruses consists of linear DNA molecules. There are three general methods of replication of DNA molecule

1. Theta (θ) mode
2. Sigma (σ) mode
3. Linear mode



The two essential functions of genetic material are replication and expression. Genetic material must replicate accurately so that progeny inherit all of the specific genetic determinants (the genotype) of the parental organism. A gene is a DNA sequence that encodes a protein, rRNA, or tRNA molecule (gene product).

Gene expression usually involves transcription of DNA into messenger RNA and translation of mRNA into protein. Genetic information encoded in DNA is expressed by synthesis of specific RNAs and proteins, and information flows from DNA to RNA to protein.

Expression of specific genetic material under a particular set of growth conditions determines the observable characteristics (phenotype) of the organism. Bacteria have few structural or developmental features that can be observed easily, but they have a vast array of biochemical capabilities and patterns of susceptibility to antimicrobial agents or bacteriophages. These latter characteristics are often selected as the inherited traits to be analyzed in studies of bacterial genetics.

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